Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS with 2+ years building **developer-facing tools**, AI workflows, full-stack products, CI/CD automation, documentation, and reliable production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **developer-facing platform workflows** for 500+ users, translating requests into APIs, validation logic, monitored jobs, technical documentation, release notes, and self-serve delivery through reviewed changes and support notes.
- Improved **backend performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with diagnostic handoffs and support runbooks.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React dashboards and Flask services** handling 10,000+ daily requests, implementing file ingestion, automated checks, agent-style AI workflows, documented fixes, and product feedback loops for users.
- Prototyped **LLM-assisted grading workflows** with OCR, model routing, cost tracking, and validation cases, reducing instructor feedback turnaround by **60%** through iterative debugging, user testing, reviews, and iteration.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **evaluation pipelines** across 100,000+ trajectory records, producing metrics, logs, reports, and reproducible review artifacts for model behavior debugging, experiment comparison, planning workflows, diagnostics, and reviews.
- Documented **edge cases and system behavior** for researchers, converting exploratory experiments into repeatable test scenarios with traceable outputs, clearer handoffs, reviewable implementation conclusions, debugging notes, and validation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **Python API and ETL services** using SQL, AWS Glue, and S3, improving response time by **30%** across integrations, release reviews, operational handoffs, troubleshooting, documentation, and supportability.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** while reducing release regressions during deployments, reviews, and checks.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **full-stack product modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, workflows, integrations, releases, debugging, validation, monitoring, support, maintenance, and reliability.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, validation notes, and shared-environment handoffs across staging workflows.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, Go familiarity, SQL, Bash, HTML/CSS
Dev Tools: Linux/Unix, shell workflows, Git, Docker, CI/CD, Jenkins, GitHub Actions, terminal-based debugging
Frontend: React, Next.js, dashboards, product interfaces, responsive UI, developer workflows, rapid prototyping
Backend: Flask, FastAPI, REST APIs, WebSockets, integrations, service design, documentation, production support
AI/Data: LLM APIs, AI-assisted coding, agent workflows, prompt evaluation, PostgreSQL, MySQL, MongoDB, observability

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL developer tool** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services, TypeScript interfaces, Docker workflows, and Git collaboration.

Big-Brain Live | *FAISS, PostgreSQL, GCP, LLM APIs*

- Engineered an **AI retrieval workbench** indexing 50K+ documents with FAISS, PostgreSQL, and GCP deployment, reducing retrieval latency by **40%** through optimized ingestion, indexing, caching, and search workflows.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ scheduling** with metric calculation and visualization, strengthening C++ fundamentals, runtime reasoning, debugging discipline, correctness testing, scheduler comparisons, and performance analysis.